

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Why Scaling is required? Discuss the various types of scaling	1	6
B) Draw the Cascode current mirror circuit and obtain its output resistance	1	6
C) Explain different types of feedback topologies in detail.	2	6
Q.2 Solve Any Two of the following.		12
A) A CS amplifier utilizes an NMOS transistor with $L = 0.36 \mu\text{m}$ and $\frac{W}{L} = 10$. It is fabricated in $0.18\mu\text{m}$ CMOS process for which $\mu_n C_{ox} = 387 \mu\text{A/V}^2$ and Early Voltage $V_A = 5 \text{ V}/\mu\text{m}$. Find the values of g_m and Intrinsic Gain A_0 for $I_D = 10 \mu\text{A}$	4	6
B) Explain MOS device capacitances in detail	3	6
C) Draw the MOSFET based Cascode amplifier and obtain mid-band frequency gain expression	5	6
Q. 3 Solve Any Two of the following.		12
A) Draw the small signal equivalent structure of differential amplifier and derive the equation for its CMRR	1	6
B) Write a short note on Wilson current mirror source	5	6
C) For Common Source amplifier let $C_{c1} = C_s = C_{c2} = 1 \mu\text{F}$, $R_G = 10\text{M}\Omega$, $R_{\text{signal}} = 100\text{K}\Omega$, $g_m = 2 \text{ mA/V}$, $R_D = R_L = R_s = 10\text{K}\Omega$ find A_m , f_{p1} , f_{p2} , f_{p3} , f_z and f_L	5	6
Q.4 Solve Any Two of the following.		12
A) Compare BJT and MOSFET current sources	1	6
B) Draw the Common Source amplifier with source degeneration resistance and obtain input and output resistances	3	6
C) Explain Miller's theorem in detail	1	6

Q. 5 Solve Any Two of the following.

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| A) Draw VI characteristics of NMOS and PMOS and explain their working | 1 | 6 |
| B) Explain Body effect in MOSFET | 4 | 6 |
| C) Derive the equation for output resistance of common gate amplifier | 4 | 6 |

***** End *****